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INDIVIDUAL DIFFERENCES IN READING SUBPROCESSES: RELATIONSHIPS BETWEEN READING ABILITY, LEXICAL ACCESS, AND EYE MOVEMENT CONTROL*

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This study examines the relationship between individual reading subprocesses and general reading ability in college students. The reading measures included eye movements while reading a passage, lexical decision latencies, comprehension, and vocabulary size. The results indicate that a distinct relation exists between reading speed and fixation behaviour associated with regressions through a text. About half the variability in comprehension scores can be predicted by subjects' performance on nonword lexical decisions, gaze durations, and vocabulary scores. These findings are discussed with reference to past studies using similar reading measures.

Keywords: reading ability, eye movement, lexical decision, comprehension, vocabulary size

INTRODUCTION

A number of theories of reading hold that reading consists of several component subprocesses that function together to produce skilled reading (see, for example, Frederiksen, 1981; LaBerge and Samuels, 1974; Wolf, 1986). The present study is an attempt to isolate a number of potential reading subprocesses (mainly those associated with accessing a word in the mental lexicon), in order to compare their variability between individuals with the variability in two measures of general reading ability, reading speed and reading comprehension. If the hypothesised subprocesses play a part in general reading ability, this will enable us to make predictions about reading ability from measures of these subprocesses, and to construct theories about the processes that underlie skilled reading.

Several tasks were used in the present study in order to provide measures of reading speed, reading comprehension, and lexical access. The first task required subjects to read silently three-line passages while their eye movements were monitored. Such monitoring

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provides information about where the eyes land within a text and for how long they stay at that location; i.e., fixation location and duration. It also provides information about the size and direction of eye movements to and from those locations (saccadic movements). The advantage of eye movement monitoring procedures is that reading behaviour can be measured without an overt verbal or manual response being made by the subject. Such responses are usually not required in silent reading, but are a component of most experimental reading tasks, such as lexical decision and naming.

Eye movement measurements have a second advantage in that they provide a means of separating different subprocesses within reading. Word identification (a subprocess that becomes active at a relatively earlier stage in processing) will presumably be reflected in "earlier" eye movement measures (e.g., initial fixations), whereas semantic integration processes (which become active at a later stage) will be reflected in "later" eye movement measures (e.g., total inspection times). Inhoff (1984), for example, found that word frequency affected initial fixations, whereas sentence context effects were apparent in gaze durations. Similar results have been found by Balota, Pollatsek, and Rayner (1985). However, Rayner and Pollatsek (1987) and O'Regan and Levy-Schoen (1987) suggested that gaze durations may also show influences of lexical processes. For these reasons, this study considered separately (i) initial fixations on a word, (ii) gaze durations (fixations on a word that occur before the eyes move outside the boundary of that word), and (iii) total fixation durations (which include fixations on a word that occur after the eyes have left its boundary, i.e., regressions).

A further measure which was obtained in this part of the study was the location of the initial fixation within certain target words in the text. These target words possess distinctive regions that distinguish them from most other words in the English language. For example, in the word "moralistic" the initial five letters are highly distinctive (there are very few other words that begin with this sequence of letters), but the final five letters are not (there are many words that end in this sequence). By contrast, a word such as "supervisor" contains a common initial sequence of letters, but a distinctive final sequence. Therefore, word identification processes would be aided if these more distinctive regions were fixated, and evidence from several sources appears to support this (Everatt and Underwood, 1992; Hyona, Niemi, and Underwood, 1989; O'Regan, Levy-Schoen, Pynte, and Brugaillere, 1984; Pynte, Kennedy, and Murray, 1991; Underwood, Clews, and Everatt, 1990). Other evidence suggests that the process of locating these more distinctive regions can occur while the eyes are still fixating a prior word. Underwood, Clews, and Everatt (1990) and Everatt and Underwood (1992) found that the first fixation within a word was sensitive to the position of the more distinctive region within that word, though this effect has not always been found (see Rayner and Morris, 1992). Given that this effect occurs at some point within the fixation data for a word (Rayner and Morris found no evidence for informative parts of a word influencing the position of the first fixation within a word, they did not discuss subsequent fixations) and that it indicates an important feature of word recognition processes (see O'Regan and Jacobs, 1992, for a discussion of this measure as an important tool to investigate word recognition), the present study investigates its relationship with reading speed and comprehension.

The second task performed by the subjects was lexical decision. This task was used in order to isolate the effects of lexical processes within reading. In a lexical decision task, all that is required is that a stimulus is compared against entries in the lexicon. If the stimulus is found in the subject's lexicon, it is a word; if it is not found, it is a nonword. This view of the lexical decision task has met with criticism in recent years. It has been suggested that the task may also involve higher level semantic processing, or could be performed on the basis of some familiarity decision which does not involve lexical access (see Neely, 1991, for a recent review). For this reason, some theorists consider a naming task more appropriate for measuring lexical access (cf. Seidenberg, Waters, Sanders, and Langer, 1984). Naming tasks, however, suffer from a similar problem. In naming, phonological output processes are necessary, and may therefore play a major role in response variability. This means that any relationships between naming latency and other reading measures will be influenced by these phonological output processes. Neither task is totally satisfactory; hence the need for a measure that removes response variability, such as eye movement data. The lexical decision task was selected for the present study because past findings indicated a sizeable relationship between lexical decisions and measures of reading ability (see Baddeley, Logie, Nimmo-Smith, and Brereton, 1985; Palmer, MacLeod, Hunt, and Davidson, 1985; Perfetti, 1983). In particular, Baddeley *et al.* (1985) argued that the relationship between lexical decision and comprehension was due to lexical access processes, a principal factor of investigation in this study.

General reading ability was measured in two ways: comprehension and reading speed. A passage reading task was used in the first part of the study to obtain eye movement data and also to determine reading speed. Each passage comprised three lines – about 20 words – of text. The average time to read these passages was used as a measure of reading speed. Subjects were also required to answer simple questions about the passages. Any subject who failed to answer these questions correctly was deemed below the required level of comprehension and was removed from the study.

A separate test, referred to as the gap test (i.e., a passage of text with single words removed, similar in structure to a cloze test), was used to measure comprehension. The subjects' task here was to fill in the gaps with the appropriate words. This could be performed correctly only if the subject comprehended the passage containing the gaps, so the number of words correctly placed in the gaps was a measure of the subjects' comprehension of the text.[1] This comprehension measure was used, rather than comprehension questions about the passages, because of inconsistent findings about the relationship between the time taken to read a passage and the number of questions about that passage correctly answered (cf. Baddeley *et al.*, 1985; Haberlandt, Graesser, and

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1 This was confirmed in a separate study using 30 subjects from a background similar to that of the present subjects. Subjects were given the same version of the gap test used here and five 60-word passages to read. After each passage they were asked four questions related to that passage. The correlation between the total number of questions correctly answered and the subjects' gap score was 0.65. The gap test also correlates (0.73) with a test of verbal ability commonly used in the UK, the Schonell Reading Test (see Pumfrey, 1985), though this involved pre-teenagers. These correlations are within the range of intercorrelations (0.64 to 0.72) found for three different measures of comprehension in Palmer *et al.* (1985). This finding, and the present results (which revealed correlations similar to those found in previous studies), suggest that the gap test is measuring comprehension.

Schneider, 1989; Palmer *et al.*, 1985; Underwood, Hubbard, and Wilkinson, 1990). These inconsistent findings may be the result of subjects in some studies using the strategy of reducing their reading speed in order to explicitly learn parts of the text for the comprehension questions, or skipping back and forth between text and comprehension questions if presented together. The gap test used here required the subjects to comprehend the text but not learn it explicitly. The passage reading task and the gap test therefore provided measures of reading speed and of comprehension, respectively. This separation of the comprehension measure from the reading speed measure removed the danger of re-reading strategies and provided a more representative measure of the relationship between processing speed and comprehension.

The primary disadvantage of utilizing the gap test was that the subject had to generate a plausible word to fit in with the rest of the passage; this required word knowledge as well as comprehension. Two factors were introduced in the tests to overcome this dual problem. Firstly, all of the missing words were common high frequency words. Secondly, a vocabulary test (a synonym judgement task) was included. If word knowledge were required in the comprehension test, then a larger than normal correlation should be found between the gap test and the vocabulary test. Past studies (for example, Baddeley *et al.*, 1985; Davis, 1968; Underwood, Hubbard, and Wilkinson, 1990) have found correlations varying from +0.3 to +0.55 between comprehension and vocabulary (although Palmer *et al.*, 1985, report larger correlations).

METHOD

Subjects

Thirty-six people drawn from the staff and students of the University of Nottingham were paid to take part in the experiment. All had normal vision.

Materials

Two pencil-and-paper tests were used in the experiment. The first was a modified version of the Mill-Hill vocabulary test, which consisted of 21 items taken mainly from the low frequency half of the Mill-Hill synonym selection test (Raven, Court, and Raven, 1977). The results were used as an index of the subjects' word knowledge. The second pencil-and-paper test was the gap test. It used the last two stories ("Brains as Machines" and "Is there Life on Mars?") from Form Y of the GAPADOL reading comprehension test (McLeod and Anderson, 1973).

For the eye movement experiment, two sets of multisyllabic words (nine or more letters) were selected. The first set contained 18 words with distinctive beginnings (signified as Ir words – informative beginnings and redundant endings), and the second set contained 18 words with distinctive endings (rI words – redundant beginnings and informative endings). These words were obtained from a pilot study in which a group of subjects different from that in the present study (but taken from the same college population) had to guess a word when only the first or last five letters were provided. The percentage of correct guesses varied due to the distribution of discriminating information (or redundancy) within the words (see Underwood, Hyona, and Niemi, 1986, and Everatt and Underwood, 1992, for a description of the type of pilot study used to obtain these words). Words with distinctive beginnings or endings were identified. The 18 words with distinctive beginning were those where the subjects were correct more than 89% of the time when given the first five letters of the word, but less than 11% of the time when given the last five letters of the word. The 18 words with distinctive ending were those where the subjects were correct less than 11% of the time when given the first five letters of the word, but more than 89% of the time when given the last five letters of the word. These words were then checked using forward and backward crossword dictionaries to ensure that they contained distinctive beginnings and endings (i.e., that there were no other words with the same beginning/ending designated as distinctive).

The words were then placed into 18 short passages. The passages were three lines long, each line containing no more than 50 character spaces. Each passage contained two target words which were positioned such that they did not appear at the beginning or end of a line nor at the beginning or end of a sentence. There were no punctuation marks around these words. Eighteen filler passages were also used. Twelve simple questions which related to the content of the passages were used to ensure that comprehension occurred. A separate set of eight passages and related questions were used as practice.

The stimuli for the lexical decision task were taken from Shapiro and Palermo (1968).[2] One hundred and eighty words were selected, containing between three and six letters. Half were converted to pronounceable nonwords by changing one letter (e.g., the word "large" was changed to "larpe", and the word "web" was changed to "wab"). Target nonwords contained, on average, the same number of letters and syllables as target words. The words used to obtain the nonwords also had, on average, word frequencies similar to those of the word targets.

Apparatus

The passages in the eye movement monitoring part of the experiment were presented on a high resolution monitor (Taxan KX-12). The subject was positioned 48 cm from the screen, by means of a chin rest. This meant that ten character spaces subtended a horizontal angle of 3 degrees, and each letter had a maximum height of 0.6 degrees. Each passage comprised three lines of text which were separated by two blank lines, each line having a maximum length of 50 character spaces, including spaces and punctuation.

[2] This source was used in order to obtain associated word pairs. Pairs with the largest association probability were used, starting with the largest probability and selecting from the list until the required number of word pairs was reached. The only exclusion criterion used was that the words should not be longer than six letters. The second word in the pair was used as a target in the lexical decision task, the first word as a prime. Prime words were then swapped between subjects so that prime/target combinations were related ("white" "black"), unrelated ("time" "black"), and neutral (the word "blank" was used as a neutral prime). Prime words were presented prior to the target words. Analyses of the results for the separate prime conditions produced correlations consistent with those obtained from reaction times averaged across prime types, which are reported here.

The eye movement recorder (Wilkinson, 1979) detected the horizontal and vertical position of the left eye by monitoring the corneal reflex from an infrared beam on a 64 × 64 photoelectric matrix. The recorder was accurate to plus or minus one character space, and sampled every 10 msec. A BBC Master computer controlled the operations of the monitor and recorded inputs from the eye movement recorder. The eye movement recorder was locked into place by means of a dental plate held between the subject's teeth onto which was mounted the recorder. Head movements were restricted by use of a chin rest and head frame. A button control held in the subject's hand was used to commence and terminate each trial.

The lexical decision stimuli were presented on an oscilloscope which was controlled by an LSI 11/23 laboratory computer. Distance from the screen and visual angle subtended by the stimuli were similar to those in the eye movement part of the experiment. Subjects responded by pressing one of two buttons situated in front of them. The computer recorded which key had been pressed and the time taken to respond.

Design and procedure

Each subject took part in the eye movement monitoring part of the experiment first. The eye movement equipment was calibrated using a central fixation point and four fixation points situated at the four corners of the viewing area. When calibration was complete, the subjects were given instructions for this part of the experiment. The subject fixated on a cursor which indicated the initial letter of a passage. The passage appeared one second later. When the subject had read the passage he/she pressed the button provided and the passage was replaced by the central calibration point. Passage reading time, which was used as a variable in the following analyses, was calculated from the time the passage appeared to the time the subject pressed the button. After renewed calibration the subject was asked to press the button for the next trial to begin.

Each subject was given eight practice passages followed by 36 experimental passages in four blocks of nine. Each passage was read quietly for comprehension. Questions relating to the passages were asked after each block to ensure that the passages had been read. Any subjects not correctly answering all but one of the questions was rejected from the study. (There was only one such case.) Rest periods were allowed after each set of questions.

After the experiment was completed, information gathered by the eye movement recorder was obtained from the BBC microcomputer. This information concerned the location of the eyes, and how long they remained at that location. Only the information around the target words was used in these analyses. They provided several measures: (i) the average length of the initial fixation on the target words; (ii) the average gaze duration; (iii) the average total fixation time; and (iv) a measure of information seeking behaviour for distinctive beginning targets (referred to as Ir words) and for distinctive ending targets (referred to as rI words).[3]

When the eye movement monitoring part of the experiment was complete, the subject was given a short break, after which the lexical decision task was performed. In total, there were 180 experimental trials in this part of the study, divided into three blocks of 60 trials. Each block was preceded by a practice set of twelve trials, and each trial was preceded by a fixation cross, which remained on the screen for 1000 msec.

Items were always centred around this cross. Target items remained on the screen until the subject responded by pressing one of the two buttons in front of them; right hand for word decisions, left for nonword decisions. After the subject responded there was an inter-trial interval of 1000 msec before the fixation cross appeared. The time to respond from the appearance of the target to the button being pressed was recorded by the computer.[4]

When the subject had completed the lexical decision task, he/she took another break, and then the two pencil-and-paper tasks were administered. In the vocabulary test, subjects were instructed to indicate which item within a list of six words meant the same as the isolated word to the left of each list (e.g., the test item was "MANUMIT" and the words to choose from were "manufacture", "liberate", "enumerate", "emanate", "accomplish", and "permit", with "liberate" being the correct answer). By way of demonstration, the first item was completed for the subjects. They were asked to guess if they did not know an item and were given as much time as they wanted. Upon completion, subjects were given the gap test, in which they were asked to fill in the gaps in the stories with words they considered appropriate. For example, the last line of the second passage was "Photographs ..... from time to time indubitable clouds which blot out temporarily large ...... of surface detail; clear weather, ... , is more usual.", and the words which complete this passage were "show" "areas" and "however". They were again allowed as much time as they required and were told to guess if they were not sure. For the vocabulary task the number of correct answers out of 20 was used as a way of assessing word knowledge. For the gap test the number of correct answers out of 37 was used as a measure of comprehension ability.

The whole experiment took between one and two hours, the main cause of variance in time being the setting up and calibration of the eye movement monitoring equipment.

RESULTS

Several variables were used in the results. Their means and standard deviations are presented in Table 1.

Relationships among the variables were established using Pearson product moment correlations (see Table 2).

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3 This measure was calculated via the position of the first fixation within these words. Fixation position was calculated as the number of character spaces from the middle of the word, with fixations to the left of centre being given a negative score. Consistent with the previous research described in the introduction, lr words were found to have initial fixations closer to the beginning of the word (mean position –1.02) than rl words (mean position – 0.10); this difference was significant, $t_{(35)} = 5.8$, $p < 0.0001$. We are using the terminology lr and rl to be consistent with the notation in previous papers (Underwood, Clews, and Everatt, 1990).

4 Errors occurred on only a small number of items; on average 1.81 (SD = 1.41) word errors and 4.53 (SD = 2.74) nonword errors per subject. Because of this highly restricted variability, errors were not included in the analyses.

TABLE 1

Means and standard deviations (*SD*) for the variables measured

Variable	Mean	*SD*	Variable	Mean	*SD*
Passage reading time (msec)	9569	3485	First fixation location: rl words (spaces)	−0.10	0.77
First fixation duration (msec)	219	32	Word lexical decision times (msec)	555	84
Gaze duration (msec)	373	73	Nonword lexical decision times (msec)	654	97
Total fixation time (msec)	568	250	Vocabulary score (words)	11.5	2.4
First fixation location: lr words (spaces)	−1.02	0.95	Gap score (comprehension) (words)	20.4	4.8

Two stepwise regression analyses were conducted. The first yielded predictors for the gap test, indicating that 51% of the variability in comprehension scores could be accounted for by three measures: Nonword lexical decision times accounted for about 35% of the variance, gaze durations accounted for another 9%, and vocabulary score for another 7%.[5]

The second regression analysis determined predictors of reading speed. A single

[5] Note that if word lexical decisions are added to nonword lexical decisions in the regression analysis, then only an extra 1% of variability in comprehension scores is explained, whereas if the alternative procedure is carried out, placing word lexical decisions into the regression first, followed by nonword decisions, the former explain 12.5% of the variability, and the latter explains an extra 23.5% of the variability. Nonword decisions appear to predict the same variability as word decisions, plus a unique amount of variability over and above that explained by word decisions. Similar results are apparent if initial fixation durations or gaze durations are added to the regression after nonword decisions. Adding gaze durations to nonword lexical decisions and initial fixation durations explains an extra 4% of variability in comprehension scores, whereas adding first fixation durations to nonword lexical decisions and gaze durations explains no extra variability.

TABLE 2

Correlations among the 10 variables identified in Table 1. Significant correlations ($p < 0.05$, two-tailed) are presented in bold. The important correlations for present purposes are those presented in the first column (passage time) and the last row (gap)

	Passage time	First dur.	Gaze	Total	Loc. lr	Loc.rl	Word RT	nonword RT	Vocab
First duration	**0.444**								
Gaze	**0.626**	**0.600**							
Total	**0.901**	**0.477**	**0.697**						
Loc. lr	−0.277	0.255	−0.251	−0.260					
Loc. rl	−0.180	−0.016	−0.146	−0.174	**0.413**				
Word RT	0.162	0.073	0.052	0.208	−0.185	−0.092			
Nonword RT	0.185	0.191	0.081	0.255	−0.002	−0.223	**0.728**		
Vocab.	0.011	0.096	−0.023	0.044	0.012	0.014	−0.150	−0.248	
Gap	−0.207	**−0.324**	**−0.347**	−0.215	0.286	0.091	**−0.354**	**−0.592**	**0.395**

variable accounted for about 80% of the variance in reading times. This was the total fixation time on the target words. When gaze durations were removed from this measure and the regression equation recalculated, the same conclusion was apparent: Total fixation times minus gaze fixation times was the only variable entered into the equation, explaining 76% of the variance in passage reading times.

DISCUSSION

We will discuss the findings of this study with reference to previous research within the literature, particularly that which used similar tasks and a similar adult subject pool.

Fixation location

The *raison d'être* for this study was to distinguish early processes within reading and investigate their relationship with general reading ability, as defined by comprehension and reading speed. The locating of distinctive regions within a word to enable faster lexical processing is such a recognition process. This measure, however, proved to have little relationship with reading speed or gap comprehension scores. In fact, the larger correlations (with lr words) suggest a relationship in the opposite direction to that expected; i.e., more negative scores (indicating fixations closer to the more informative beginning of the word) are related to slower reading speeds and lower gap comprehension scores. There is also little correlation with fixation duration, a measure that must, to some extent, be related to lexical access processes. Either the present measure did not capture the information seeking behaviour of the subjects,[6] or it was not related to lexical access. Our own view of this information seeking effect is that it is an aid to lexical access (see also O'Regan and Jacobs, 1992). If the most informative parts of a word can be located by an eye movement, then lexical access should be easier. Therefore, we expected those who show more information seeking behaviour to be better readers due to their improved lexical access procedures. Our finding of little or no relationship between information seeking and measures of speed and comprehension do not support this view. If any relationship exists, the evidence from words with informative beginnings suggests that increased information seeking behaviour is related to *poorer* speed and comprehension performance. Future research on information seeking behaviour (cf. Underwood, Clews, and Everatt, 1990; Everatt and Underwood, 1992) or convenient/optimal viewing positions (cf. O'Regan and Levy-Schoen, 1987; O'Regan and Jacobs, 1992) will have to take account of this finding.

Fixation duration

The fixation duration data (total, gaze, and initial fixation data) showed significant correlations with passage reading time. However, the main predictor of reading speed is total fixating time, suggesting that the amount of time spent re-fixating a target word is the main predictor of passage reading time (we tested this directly by removing gaze durations times from the total fixation data and found an equally large correlation with passage reading time, $r = 0.87$). This indicates that the more a subject re-reads the targets, the slower their reading of the whole passage becomes. Similar findings are apparent in the data of Underwood, Hubbard, and Wilkinson (1990) with reading speed being very closely linked to the number of regressions ($r = 0.83$) within a text.

Fixation duration also showed relationships with the comprehension test. Both first fixation durations and gaze durations were significantly correlated with gap scores, suggesting that shorter initial and gaze fixations are related to larger comprehension scores. Total fixation durations, however, do not appear to show such a relationship. Re-reading of the text is less related to reading comprehension than it is to reading speed. Again, these findings are consistent with previous results in the literature. Underwood, Hubbard, and Wilkinson (1990) also found no significant correlations between comprehension and the number of regressions and number of fixations within a text, but

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[6] O'Regan and Levy-Schoen (1987) argue that the second fixation is sensitive to distinctive regions of the text. It is possible that the position of the second fixation should have been the variable used in the present analyses, and indeed there was a tendency for second fixations on rI words to be nearer the end of the word compared to that for Ir words (0.39 compared to 0.07; remember that a larger positive value indicates a position further to the right of centre). Correlations between the position of the second fixation and reading speed were only slightly larger than those for the position of the first fixation ($r = -0.3$ for Ir words and $r = -0.26$ for rI words); and correlations with comprehension scores were similarly smaller than for reading speed ($r = 0.04$ with Ir words and $r = 0.22$ with rI words). Thus, more or less the same pattern of results was apparent when the first fixation position was used as the dependent variable.

larger correlations between comprehension and average fixation duration (around $r = -0.4$).

In the regression analysis, gaze duration appeared to be the second largest predictor of variability in comprehension scores, after nonword lexical decisions. Although first fixation durations also correlated with gap scores, gaze durations explained a unique amount of variability over and above that explained by initial fixation durations. Whether this unique variability is involved in processes after initial recognition or in recognition itself remains to be determined.

The findings from the present study and those of Underwood, Hubbard, and Wilkinson (1990) suggest that both speed and comprehension can be predicted to some extent by fixation duration data, indicating that such duration data can be a useful tool in predicting speed and comprehension performance. The interesting aspect, however is that speed and comprehension are related to different duration measures. Comprehension appears to be related to duration measures early within text processing, perhaps to those processes involved in speed of word identification, whereas reading speed is related to duration measures late within text processing, perhaps to text integration processes.

Reading speed and comprehension

In concurrence with previous studies, we found no evidence for a relationship between reading speed and comprehension (see also Underwood, Hubbard, and Wilkinson, 1990; Jackson and McClelland, 1979; Baddeley *et al.*, 1985; although the correlations found by Palmer *et al.*, 1985, were larger than those of the present study they too concluded that reading speed and comprehension should be treated as "distinct abilities", p. 82). A novel feature of the present study was the attempt to reduce re-reading strategies by using separate measures of reading speed and comprehension. The reasoning being that such re-reading strategies may mask any underlying relationship between comprehension and speed. However, there was still little evidence for a relationship between speed and comprehension. The results of the present study indicate that if speed is of importance within comprehension, it is the speed of processing a single word, rather than a whole passage.

Lexical decision latencies

The most surprising results in the present study come from the lexical decision task. They suggest that lexical decision time is related to comprehension but not reading speed, and that nonword decisions show a larger correlation with comprehension than word decisions. Past studies have also found a relationship between comprehension and lexical decisions. Baddeley *et al.* (1985) found a significant correlation between comprehension ability and lexical decision times, and a negligible correlation between lexical decision and reading speed. However, they did not separate word and nonword decisions. If we average our word and nonword data, the correlation with comprehension is very similar to that of Baddeley *et al.*

Why is there a larger relationship between comprehension and nonword decisions compared to that between comprehension and word decisions? Possibly word lexical decisions may show smaller variability in response times since the words were, in the main, high frequency words, which may be accessed at a similar rate by all subjects. Nonword responses may act like low frequency words and require more processing within an individual's lexicon. Here variability in performance will be apparent across subjects thus producing a larger correlation. A similar argument was put forward by Schwartz (1984) when describing two studies comparing the GAPADOL test (the same test as used in the present study) and measures of word and pseudoword naming. As in the present study larger relationships were found between comprehension and pseudoword performance compared to that between comprehension and word performance; though correlations were smaller in the naming task compared to the lexical decision task of the present study. Schwartz argued that the higher correlation for pseudowords was due to larger variability in their processing.

However, the lack of a relationship between lexical decision latencies and fixation durations suggests that the lexical access explanation may not be correct. The variability in both lexical decision and fixation duration measures cannot simply be due to lexical access since a larger relationship should then be found.[7] An alternative explanation is that the lexical decision task not only involves lexical access, but also involves decisions (see, for example, the discussion of how familiarity decisions can contaminate the lexical decision task by Besner and Johnston, 1989). Decisions are also involved in comprehension tasks. In the present study, the gap test involves decisions about which of several possible alternatives should be used to complete a passage. Decisions within each task will involve higher order meta-cognitive processes such as an analysis of the task, estimations of probabilities of an answer being correct, manipulation of alternative answers, etc. Relationships between comprehension and lexical decisions may stem, or be increased, by these common decision processes.[8] This account would explain the reduced relationship between nonwords and comprehension when naming is required (Schwartz, 1984). Fewer decision processes may be required with naming, perhaps simply whether to name by analogy or whether to rely on spelling-sound correspondences (see Ehri, 1991, for a discussion of these two methods). Therefore, in studies of lexical processing we propose that eye movement monitoring be utilised in place of, or in addition to, the lexical decision task. Alternatively, tasks which "partial out" the effects of decision processes should be used in conjunction with the lexical decision task.

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7 An alternative explanation for the small correlation between fixation duration and lexical decision latency is that the lexical decision task used short, high frequency words, whereas the eye movement data were derived from target words that were low frequency and much longer. However, in a separate study using 25 subjects from the same population as the present study, lexical decisions for short, high frequency words and for long, low frequency words (similar to those used in the present study) correlated quite highly ($r = 0.7$). This suggests that the processes that go into lexical decisions for both types of words are similar, and does not account for the low correlation between eye movement data and lexical decision data in the present study. Also, if the correlation between lexical decision times and average fixations on short, high frequency words within the passages in the present study is calculated, we again find a small relationship between lexical decisions and fixation duration ($r = 0.24$).

8 A similar explanation has been proposed for the relationship between semantic encoding tasks and reading ability (Whitney, Kellas, and Ferraro, 1990).

Conclusion

In conclusion, the present results suggest that (at least within a college population) there is little relationship between passage reading speed and reading comprehension. This study concurs with the conclusion of Palmer *et al.* (1985) that it would be better to consider reading speed and reading comprehension as separate measures. This conclusion is supported by the finding of separate predictors of comprehension and reading speed within the eye movement data collected; basically, increased re-fixation behaviour is related to slower reading speeds, and shorter gaze durations are associated with better comprehension. The present findings therefore support those of Underwood, Hubbard, and Wilkinson (1990): Eye movement data can be informative as to individual differences in reading ability. We also suggest that eye movement measurement may be preferable to more traditional methods (e.g., the lexical decision task) when studying such individual differences. Apropos the lexical decision data, the results of the present study indicate a strong relationship between comprehension and nonword lexical decisions. The ability to reject items in this task appears to be related to the ability to comprehend a piece of text. Further investigations will be necessary, however, to discover the cause of this relationship.

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